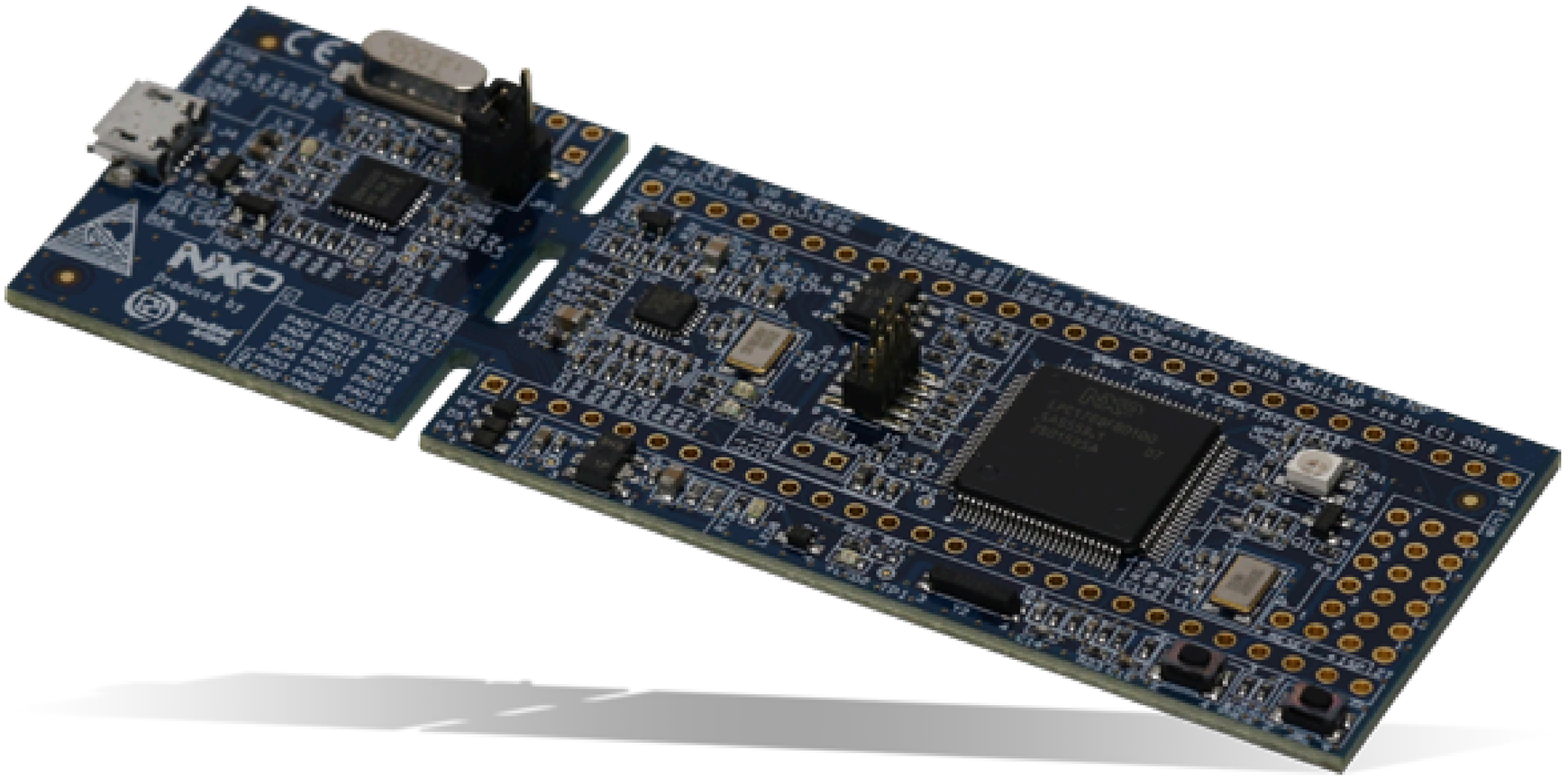
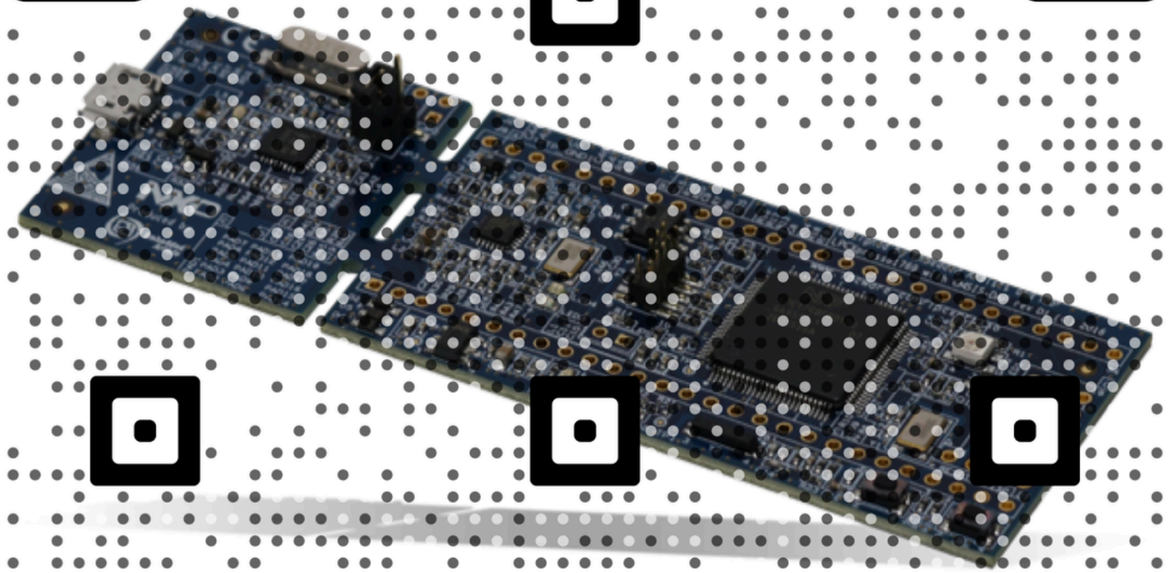


ADC

David Trujillo



ADC

AVAILABLE PINS

AD 0	P0.23	01
AD 1	P0.24	01
AD 2	P0.25	01
AD 3	P0.26	01
AD 4	P1.30	11
AD 5	P1.31	11
AD 6	P0.3	10
AD 7	P0.2	10

CLOCK CONTROL

Table 46. Power Control for Peripherals register (PCONP - address 0x400F C0C4) bit description

Bit	Symbol	Description	Reset value
12	PCADC	A/D converter (ADC) power/clock control bit. **Note:** Clear the PDN bit in the AD0CR before clearing this bit, and set this bit before setting PDN.	0

Table 40. Peripheral Clock Selection register 0 (PCLKSEL0 - address 0x400F C1A8) bit description

Bit	Symbol	Description	Reset value
25:24	PCLK_ADC	Peripheral clock selection for ADC.	00

Table 42. Peripheral Clock Selection register bit values

PCLKSEL0 and PCLKSEL1 individual peripheral's clock select options	Function	Reset value
00	PCLK_peripheral = CCLK/4	00
01	PCLK_peripheral = CCLK	
10	PCLK_peripheral = CCLK/2	
11	PCLK_peripheral = CCLK/8, except for CAN1, CAN2, and CAN filtering when "11" selects = CCLK/6.	

PIN CONFIGURATION

Table 531. ADC registers

Generic Name	Description	Access	Reset value[1]	AD0 Name & Address
ADCR	A/D Control Register. The ADCR register must be written to select the operating mode before A/D conversion can occur.	R/W	1	AD0CR - 0x4003 4000
ADGDR	A/D Global Data Register. This register contains the ADC's DONE bit and the result of the most recent A/D conversion.	R/W	NA	AD0GDR - 0x4003 4004
ADINTEN	A/D Interrupt Enable Register. This register contains enable bits that allow the DONE flag of each A/D channel to be included or excluded from contributing to the generation of an A/D interrupt.	R/W	0x100	AD0INTEN - 0x4003 400C
ADDR0	A/D Channel 0 Data Register. This register contains the result of the most recent conversion completed on channel 0.	RO	NA	AD0DR0 - 0x4003 4010
ADDR1	A/D Channel 1 Data Register. This register contains the result of the most recent conversion completed on channel 1.	RO	NA	AD0DR1 - 0x4003 4014
ADDR2	A/D Channel 2 Data Register. This register contains the result of the most recent conversion completed on channel 2.	RO	NA	AD0DR2 - 0x4003 4018
ADDR3	A/D Channel 3 Data Register. This register contains the result of the most recent conversion completed on channel 3.	RO	NA	AD0DR3 - 0x4003 401C
ADDR4	A/D Channel 4 Data Register. This register contains the result of the most recent conversion completed on channel 4.	RO	NA	AD0DR4 - 0x4003 4020
ADDR5	A/D Channel 5 Data Register. This register contains the result of the most recent conversion completed on channel 5.	RO	NA	AD0DR5 - 0x4003 4024
ADDR6	A/D Channel 6 Data Register. This register contains the result of the most recent conversion completed on channel 6.	RO	NA	AD0DR6 - 0x4003 4028
ADDR7	A/D Channel 7 Data Register. This register contains the result of the most recent conversion completed on channel 7.	RO	NA	AD0DR7 - 0x4003 402C
ADSTAT	A/D Status Register. This register contains DONE and OVERRUN flags for all of the A/D channels, as well as the A/D interrupt/DMA flag.	RO	0	AD0STAT - 0x4003 4030
ADTRM	ADC trim register.	R/W	0x0000 0F00	AD0TRM - 0x4003 4034

Table 532: A/D Control Register (AD0CR - address 0x4003 4000) bit description

Bit	Symbol	Value	Description	Reset value
7:0	SEL		Selects which of the AD0.7:0 pins is (are) to be sampled and converted. For AD0, bit 0 selects Pin AD0.0, and bit 7 selects pin AD0.7. In software-controlled mode, only one of these bits should be 1. In hardware scan mode, any value containing 1 to 8 ones is allowed. All zeroes is equivalent to 0x01.	0x01
15:8	CLKDIV		The APB clock (PCLK_ADC0) is divided by (this value plus one) to produce the clock for the A/D converter, which should be less than or equal to 13 MHz. Typically, software should program the smallest value in this field that yields a clock of 13 MHz or slightly less, but in certain cases (such as a high-impedance analog source) a slower clock may be desirable.	0
16	BURST	1	The AD converter does repeated conversions at up to 200 kHz, scanning (if necessary) through the pins selected by bits set to ones in the SEL field. The first conversion after the start corresponds to the least-significant 1 in the SEL field, then higher numbered 1-bits (pins) if applicable. Repeated conversions can be terminated by clearing this bit, but the conversion that's in progress when this bit is cleared will be completed. **Remark:** START bits must be 000 when BURST = 1 or conversions will not start. If BURST is set to 1, the ADGINTEN bit in the AD0INTEN register (Table 534) must be set to 0.	0
		0	Conversions are software controlled and require 65 clocks.	

Table 532: A/D Control Register (AD0CR - address 0x4003 4000) bit description

Bit	Symbol	Value	Description	Reset value
21	PDN	1	The A/D converter is operational.	0
		0	The A/D converter is in power-down mode.	
23:22	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
26:24	START		When the BURST bit is 0, these bits control whether and when an A/D conversion is started:	0
		000	No start (this value should be used when clearing PDN to 0).	
		001	Start conversion now.	
		010	Start conversion when the edge selected by bit 27 occurs on the P2.10 / EINT0 / NMI pin. Set the pin function to EINT0 in PINSEL4 register.	
		011	Start conversion when the edge selected by bit 27 occurs on the P1.27 / CLKOUT / USB_OVRCRn / CAP0.1 pin. Set the pin function to CAP0.1 in PINSEL3 register.	
		100	Start conversion when the edge selected by bit 27 occurs on MAT0.1. Note that this does not require that the MAT0.1 function appear on a device pin.	
		101	Start conversion when the edge selected by bit 27 occurs on MAT0.3. Note that it is not possible to cause the MAT0.3 function to appear on a device pin.	
		110	Start conversion when the edge selected by bit 27 occurs on MAT1.0. Note that this does not require that the MAT1.0 function appear on a device pin.	
		111	Start conversion when the edge selected by bit 27 occurs on MAT1.1. Note that this does not require that the MAT1.1 function appear on a device pin.	
27	EDGE		This bit is significant only when the START field contains 010-111. In these cases:	0
		1	Start conversion on a falling edge on the selected CAP/MAT signal.	
		0	Start conversion on a rising edge on the selected CAP/MAT signal.	

Table 533: A/D Global Data Register (AD0GDR - address 0x4003 4004) bit description

Bit	Symbol	Description	Reset value
3:0	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
15:4	RESULT	When DONE is 1, this field contains a binary fraction representing the voltage on the AD0[n] pin selected by the SEL field, as it falls within the range of V_{REFP} to V_{REFN}. Zero in the field indicates that the voltage on the input pin was less than, equal to, or close to that on V_{REFN}, while 0xFFF indicates that the voltage on the input was close to, equal to, or greater than that on V_{REFP}.	NA
23:16	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
26:24	CHN	These bits contain the channel from which the RESULT bits were converted (e.g. 000 identifies channel 0, 001 channel 1...).	NA
29:27	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
30	OVERRUN	This bit is 1 in burst mode if the results of one or more conversions was (were) lost and overwritten before the conversion that produced the result in the RESULT bits. This bit is cleared by reading this register.	0
31	DONE	This bit is set to 1 when an A/D conversion completes. It is cleared when this register is read and when the ADCR is written. If the ADCR is written while a conversion is still in progress, this bit is set and a new conversion is started.	0

Table 534: A/D Interrupt Enable register (AD0INTEN - address 0x4003 400C) bit description

Bit	Symbol	Value	Description	Reset value
0	ADINTEN0	0	Completion of a conversion on ADC channel 0 will not generate an interrupt.	0
		1	Completion of a conversion on ADC channel 0 will generate an interrupt.	
			⋮	
7	ADINTEN7	0	Completion of a conversion on ADC channel 7 will not generate an interrupt.	0
		1	Completion of a conversion on ADC channel 7 will generate an interrupt.	
8	ADGINTEN	0	Only the individual ADC channels enabled by ADINTEN7:0 will generate interrupts. **Remark:** This bit must be set to 0 in burst mode (BURST = 1 in the AD0CR register).	1
		1	Only the global DONE flag in ADDR is enabled to generate an interrupt.	

Table 535: A/D Data Registers (AD0DR0 to AD0DR7 - 0x4003 4010 to 0x4003 402C) bit description

Bit	Symbol	Description	Reset value
3:0	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
15:4	RESULT	When DONE is 1, this field contains a binary fraction representing the voltage on the AD0[n] pin, as it falls within the range of V_{REFP} to V_{REFN}. Zero in the field indicates that the voltage on the input pin was less than, equal to, or close to that on V_{REFN}, while 0xFFF indicates that the voltage on the input was close to, equal to, or greater than that on V_{REFP}.	NA
29:16	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
30	OVERRUN	This bit is 1 in burst mode if the results of one or more conversions was (were) lost and overwritten before the conversion that produced the result in the RESULT bits.This bit is cleared by reading this register.	
31	DONE	This bit is set to 1 when an A/D conversion completes. It is cleared when this register is read.	NA

Table 536: A/D Status register (AD0STAT - address 0x4003 4030) bit description

Bit	Symbol	Description	Reset value
0	DONE0	This bit mirrors the DONE status flag from the result register for A/D channel 0.	0
		⋮	
7	DONE7	This bit mirrors the DONE status flag from the result register for A/D channel 7.	0
8	OVERRUN0	This bit mirrors the OVERRRUN status flag from the result register for A/D channel 0.	0
		⋮	
15	OVERRUN7	This bit mirrors the OVERRRUN status flag from the result register for A/D channel 7.	0
16	ADINT	This bit is the A/D interrupt flag. It is one when any of the individual A/D channel Done flags is asserted and enabled to contribute to the A/D interrupt via the ADINTEN register.	0
31:17	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

Table 537: A/D Trim register (ADTRM - address 0x4003 4034) bit description

Bit	Symbol	Description	Reset value
3:0	-	reserved.	NA
7:4	ADCOFFS	Offset trim bits for ADC operation. Initialized by the boot code. Can be overwritten by the user.	0
11:8	TRIM	written-to by boot code. Can **not** be overwritten by the user. These bits are locked after boot code write.	1111
31:12	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

```c
typedef struct {
  __IO uint32_t ADCR;
  __IO uint32_t ADGDR;
       uint32_t RESERVED0;
  __IO uint32_t ADINTEN;
  __I  uint32_t ADDR0;
  __I  uint32_t ADDR1;
  __I  uint32_t ADDR2;
  __I  uint32_t ADDR3;
  __I  uint32_t ADDR4;
  __I  uint32_t ADDR5;
  __I  uint32_t ADDR6;
  __I  uint32_t ADDR7;
  __I  uint32_t ADSTAT;
  __IO uint32_t ADTRM;
} LPC_ADC_TypeDef;
```

LPC_ADC →

DRIVERS

```c
typedef enum {
    ADC_CHANNEL_0 = 0,
    ADC_CHANNEL_1,
    ADC_CHANNEL_2,
    ADC_CHANNEL_3,
    ADC_CHANNEL_4,
    ADC_CHANNEL_5,
    ADC_CHANNEL_6,
    ADC_CHANNEL_7
} ADC_CHANNEL;
```

```c
typedef enum {
    ADC_INT_CH0
    ADC_INT_CH1
    ADC_INT_CH2
    ADC_INT_CH3
    ADC_INT_CH4
    ADC_INT_CH5
    ADC_INT_CH6
    ADC_INT_CH7
    ADC_INT_GLO
} ADC_INT_SOURCE;
```

```c
typedef enum {
    ADC_START_CONTINUOUS = 0,
    ADC_START_NOW,
    ADC_START_ON_EINT0,
    ADC_START_ON_CAP01,
    ADC_START_ON_MAT01,
    ADC_START_ON_MAT03
} ADC_START_MODE;
```

```
typedef enum {
    ADC_START_ON_RISING = 0,
    ADC_START_ON_FALLING
```

```
typedef enum {
    ADC_DATA_OVERRUN = 0,
    ADC_DATA_DONE
} ADC_DATA_STATUS;
```

```
void ADC_Init(uint32_t rate)
```

Initializes the ADC peripheral with a specific sampling rate.

This function enables the ADC clock in the `PCONP` register, calculates the required clock divider based on the peripheral clock (`PCLK`) to achieve the target rate, and powers up the converter.

Params: **rate** — Desired sampling rate in Hz (e.g., 100000 for 100kHz).

Remarks: A full conversion requires 65 ADC clock cycles. The maximum supported conversion rate for the LPC176x is 200 kHz.

`lpc17xx_adc.h`

`void ADC_DeInit()`

De-initializes the ADC peripheral.

Powers down the ADC by clearing the `PDN` bit in the `ADCR` register and disables the peripheral clock to minimize power consumption.

`lpc17xx_adc.h`

```
void ADC_PinConfig(ADC_CHANNEL channel)
```

Configures the physical pin associated with an ADC channel.

Sets the corresponding `PINSEL` register to enable the analog input function for the selected channel and configures the `PINMODE` register to "Neither pull-up nor pull-down" (Tristate) to prevent measurement interference.

Params: **channel** — ADC channel to be configured (`ADC_CHANNEL_x` [0...7]).

`lpc17xx_adc.h`

void ADC_BurstEnable()

Enables the ADC Burst conversion mode.

In Burst mode, the ADC performs repeated conversions on all selected channels. It is highly recommended for applications requiring continuous monitoring of multiple analog signals without manual software triggering.

Remarks:

- `START` bits must be `000` before enabling burst mode.
- `ADGINTEN` bit must be disabled for burst mode operation.
- Burst mode starts conversions immediately.

`lpc17xx_adc.h`

```
void ADC_BurstDisable()
```

Disables the ADC Burst conversion mode.

Switches the ADC back to software-controlled mode. Conversions will only occur when explicitly triggered via a start command or external trigger.

Remarks:

- Disabling burst mode stops conversions.

```
void ADC_PowerUp()
```

Powers up the ADC converter.

Sets the `PDN` bit in the `ADCR` register, bringing the ADC out of power-down mode to its operational state.

Remarks:

- Enable only after enabling the ADC clock.
- Does not enable the ADC clock.

`lpc17xx_adc.h`

```
void ADC_PowerDown()
```

Powers down the ADC converter.

Clears the `PDN` bit in the `ADCR` register. This reduces the power consumption of the peripheral.

Remarks:

- Disable before disabling the ADC clock.
- Does not disable the ADC clock.

```
void ADC_StartCmd(ADC_START_MODE mode)
```

Configures the start mode or trigger for the ADC conversion.

This function clears the previous start bits and sets the new trigger mode. It supports starting the conversion immediately (Software Start) or synchronizing it with external hardware signals such as `EINT0`, `CAP`, or `MAT.`

Params: **mode** — Selected trigger source:

- `ADC_START_CONTINUOUS` (burst mode)
- `ADC_START_NOW` (manual start)
- `ADC_START_ON_EINT0`
- `ADC_START_ON_CAP01`
- `ADC_START_ON_MAT01`
- `ADC_START_ON_MAT03`

```
void ADC_ChannelEnable(ADC_CHANNEL channel)
```

Selects an ADC channel for the next conversion sequence.

Adds the specified channel to the bitmask of active channels in the `ADCR` register. Multiple channels can be enabled simultaneously, especially useful in Burst mode.

Params: **channel** — The ADC channel to enable (`ADC_CHANNEL_x` [0...7]).

`lpc17xx_adc.h`

```
void ADC_ChannelDisable(ADC_CHANNEL channel)
```

De-selects an ADC channel from the conversion sequence.

Removes the specified channel from the active bitmask. If no channels remain selected, the ADC will select the first channel (`AD0.0`) by default.

Params: **channel** — The ADC channel to disable (`ADC_CHANNEL_x` [0...7]).

`lpc17xx_adc.h`

```
void ADC_EdgeStartConfig(ADC_START_ON_EDGE edge)
```

Configures the trigger edge for external hardware starts.

Determines whether the ADC conversion starts on the rising or falling edge of the selected hardware trigger (e.g., `CAP` or `MAT` signals). This setting is only relevant when the start mode is not set to "Now".

Params: **edge** — The desired edge (`ADC_START_ON_RISING` or `ADC_START_ON_FALLING`).

Remarks:

- This option is only effective when the start bits are set to [`010` ... `111`]

`lpc17xx_adc.h`

```
void ADC_IntEnable(ADC_INT_SOURCE source)
```

Enables an interrupt source for the ADC.

Configures the `ADINTEN` register to generate an interrupt when a specific channel completes its conversion or when the global `DONE` flag is set.

Params: **source** — The interrupt source to enable (`ADC_INT_CHx` [`0` ... `7`] or `ADC_INT_GLOBAL`).

Remarks:

- If `ADC_INT_GLOBAL` is selected, only the global DONE flag is enabled to generate an interrupt.
- Do not enable `ADC_INT_GLOBAL` if burst mode is enabled.

`lpc17xx_adc.h`

```
void ADC_IntDisable(ADC_INT_SOURCE source)
```

Disables an interrupt source for the ADC.

Clears the corresponding bit in the `ADINTEN` register, preventing the peripheral from generating interrupt requests for the specified source.

Params: **source** — The interrupt source to disable (`ADC_INT_CHx` [`0` ... `7`] or `ADC_INT_GLOBAL`).

`lpc17xx_adc.h`

```
FlagStatus ADC_GlobalGetStatus(ADC_DATA_STATUS type)
```

Reads the global status of the most recent ADC conversion.

Checks the Global Data Register (`ADGDR`) to determine if a conversion is complete or if an overrun condition has occurred across any channel.

Params: **type** — Status type to check (`ADC_DATA_DONE` or `ADC_DATA_OVERRUN`).

Returns: `SET` if the flag is active, `RESET` otherwise.

`lpc17xx_adc.h`

```
FlagStatus ADC_ChannelGetStatus(
    ADC_CHANNEL channel,
    ADC_DATA_STATUS type)
```

Reads the status of a specific ADC channel.

Accesses the individual channel data register (`ADDR0` - `ADDR7`) to check the completion or overrun status for that specific input.

Params: **channel** — The ADC channel to check (`ADC_CHANNEL_x` [0...7]).
type — Status type (`DONE` or `OVERRUN`).

Returns: `SET` if the flag is active, `RESET` otherwise.

`lpc17xx_adc.h`

```
uint16_t ADC_GlobalGetData()
```

Retrieves the result of the last conversion from any channel.

Returns the 12-bit digital value stored in the Global Data Register.

Returns: The 12-bit conversion result (`0` to `4095`).

Remarks:

- The returned value is right-aligned to bits [`11`:`0`].

```
uint16_t ADC_ChannelGetData(ADC_CHANNEL channel)
```

annel.

Retrieves the conversion result for a specific ADC channel.

Returns the 12-bit digital value from the dedicated register of the selected channel.

Params: **channel** — The ADC channel to read (`ADC_CHANNEL_x` [`0` ... `7`]).

Returns: The 12-bit conversion result (`0` to `4095`).

Remarks:

- The returned value is right-aligned to bits [`11`:`0`].
- Reading this register automatically clears the `DONE` flag for that specific channel.

`lpc17xx_adc.h`